

CZ/CE2006 S1 AY2018-19

Past Exam Questions Practice

The following set of exam questions (Q3 and Q4) were from past few semesters. Some of these questions have been modified slightly for this semester exam questions practice during the lecture.

Dr. Sim's sample solution guides to Q3 and Q4 will be discussed during the last two lectures. The sample solution guides will not be posted on NTULearn.

Black Box Testing:

- Equivalence Class (EC) and
- Boundary Value (BV) Testing

4. (a) The XYZ Bank supports a couple to buy a home under the following two approval conditions:

C1: The couple must be currently employed full time.

C2: The couple current combined annual total income must not be more than \$100,000 (inclusive). Specifically, the couple combined annual total income must be between \$0 (inclusive) and \$99,999 (inclusive). The income must be an integer value.

- (i) Determine the **equivalence classes** for the above two approval conditions.
(3 marks)
- (ii) Determine the **boundaries** of the equivalence classes identified in your answers to Q4(a)(i). For each boundary, identify a value **on the boundary**, a value **just below** the boundary, and a value **just above** the boundary.
(4 marks)
- (iii) You intend to perform **defensive testing** of the local access option. Design a set of **test cases to test the two above approval conditions based on the equivalence classes and boundary values** identified in your answers to Q4(a)(i) and Q4(a)(ii).
(5 marks)

White Box Testing:

Control Flow Graph (CFG), Cyclomatic Complexity (CC), and Basis Path Testing

4.(b) When the user selects a position on Google maps, the CloudPSI application determines the closest city within 10 kilometers of the selected position. The *findCity(List<City> cities, Position position)* method (as shown in the Java code in Figure Q4) implements this application logic.

(i) Draw the **control flow graph** for the *findCity(List<City> cities, Position position)* method.
[Use line numbers on process blocks and decision points for clarity]
(4 marks)

(ii) Calculate the **Cyclomatic Complexity** of the *findCity(List<City> cities, Position position)* method.
(2 marks)

(iii) List the **set of basis paths** for performing basis path testing of the *findCity(List<City> cities, Position position)* method. Design a test case (including the **input parameter List<City>**, the **distance** between the input position and a city's position, and the **expected return value**) for each of the basis paths.
(8 marks)


```
1.    City findCity(List<City> cities, Position position) {  
2.        City city = null;  
3.        double shortestDistance = MAX_DOUBLE;  
4.        for(int i = 0; i < cities.size(); i++) {  
5.            double distance = computeDistance(cities[i].position, position);  
6.            if(distance <= 10) {  
7.                if(distance < shortestDistance) {  
8.                    city = cities[i];  
9.                    shortestDistance = distance;  
10.                }  
11.            }  
12.        }  
13.        return city;  
14.    }
```

Figure Q4

Architecture and Design Patterns

3. (a) Refer to the classes identified in your answer to Q2(a).

- (i) Apply and draw the **Layered Architecture** to model these classes. Indicate the **boundary class**, **control class**, and **entity class** on the layered architecture diagram. Indicate or state clearly the **dependencies** between the layers and/or classes on the layered architecture diagram.

[You may use symbols to indicate the boundary class, control class, and entity class] (6 marks)

- (ii) List **THREE strengths** of layered architecture. List **ONE weakness** of layered architecture. (4 marks)

- (iii) Use ONE to TWO sentences to **explain briefly the core responsibility of boundary class, control class, and entity class.**

(3 marks)

- 3.(b) Currently, borrowers from XYZ Bank return their borrowed money via three payment methods: NETs, Visa, or MasterCard. Payment methods can be expanded in the future to accommodate borrowers' needs and new payment methods, such as online payment (i.e. Paypal) and mobile payment.
- (i) Identify and explain briefly **TWO (2) design problems** to implement the payment methods requirement. (3 marks)
 - (ii) Propose an appropriate **design pattern to address the design problems** identified in your answer in Q3(b)(i). (2 marks)
 - (iii) Depict the application of the design pattern proposed in Q3(b)(ii) in a **Class Diagram**. Explain briefly the **roles** each class plays in the proposed design pattern. (8 marks)